

Objective 1:

To simulate an FDMA system by allocating individual frequency channels to multiple users

and determine the channel bandwidth assigned to each user.

```
clc;
```

```
clear;
```

```
close all;
```

```
% Sampling Frequency
```

```
fs = 1000;
```

```
t = 0:1/fs:1;
```

```
% FDMA Parameters
```

```
TBW = 600;    % Total Bandwidth (kHz)
```

```
Users = 3;    % Number of Users
```

```
CBW = TBW / Users; % Bandwidth per User
```

```
% Carrier Frequencies
```

```
fc = [100 300 500];
```

```
% Message Signal
```

```
m = sin(2*pi*10*t);
```

```
% Modulated Signals
```

```
s1 = m .* cos(2*pi*fc(1)*t);
```

```
s2 = m .* cos(2*pi*fc(2)*t);
```

```
s3 = m .* cos(2*pi*fc(3)*t);
```

```

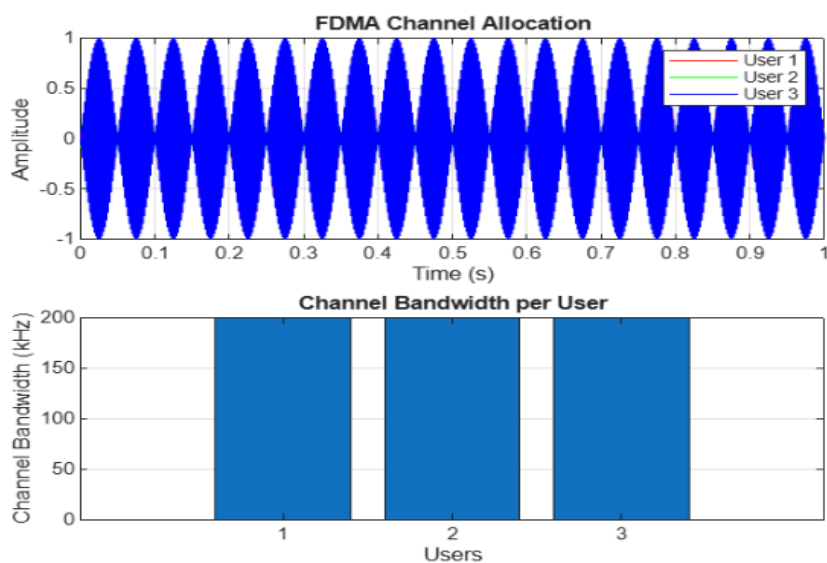
% Plot Modulated Signals
subplot(2,1,1);
plot(t, s1, 'r', t, s2, 'g', t, s3, 'b');
grid on;
title('FDMA Channel Allocation');
xlabel('Time (s)');
ylabel('Amplitude');
legend('User 1', 'User 2', 'User 3');

```

```

% Plot Bandwidth Allocation
subplot(2,1,2);
bar(1:Users, CBW * ones(1, Users));
grid on;
xlabel('Users');
ylabel('Channel Bandwidth (kHz)');
title('Channel Bandwidth per User');

```



Objective 2:

To calculate the data rate achieved by each user in the FDMA system based on the allocated

channel bandwidth.

Output Parameter: Data Rate (kbps)

```
clc;
```

```
clear;
```

```
close all;
```

```
% FDMA Parameters
```

```
BW = 200;    % Channel Bandwidth (kHz)
```

```
Users = 3;    % Number of Users
```

```
M = 2;        % BPSK Modulation
```

```
% Data Rate Calculation
```

```
DataRate = BW * log2(M); % Data Rate (kbps)
```

```
Rate = DataRate * ones(Users,1);
```

```
% Display Results
```

```
disp('User Data Rate (kbps)');
```

```
disp(table((1:Users)', Rate, ...
```

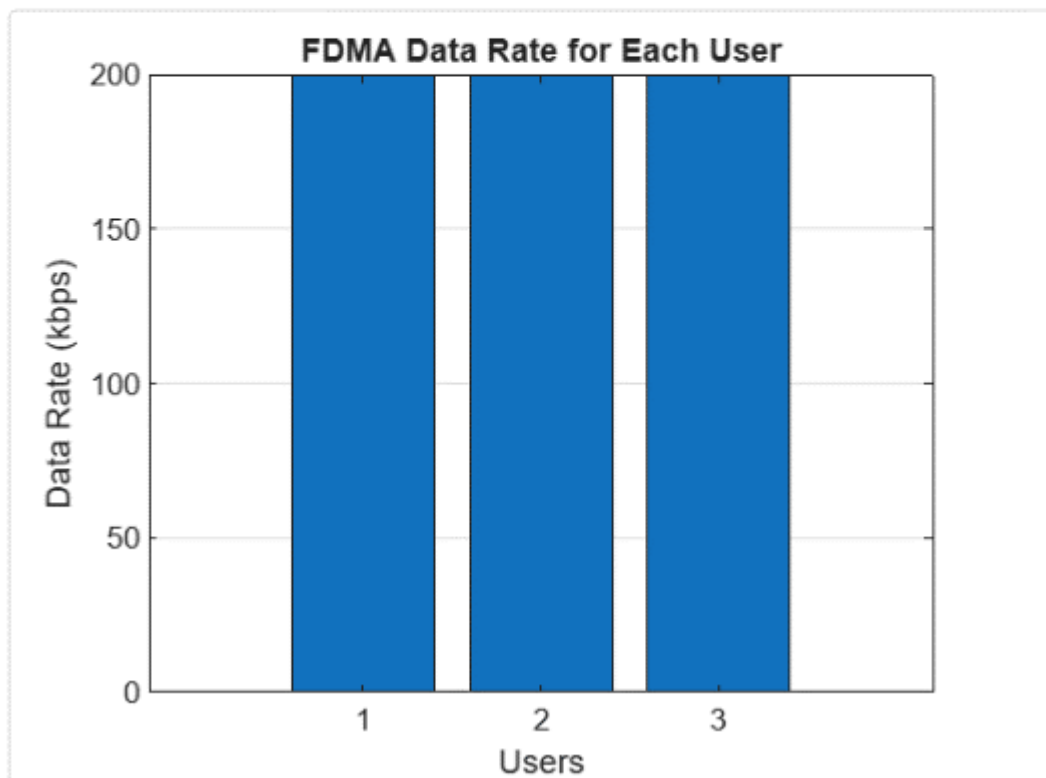
```
    'VariableNames', {'User', 'DataRate_kbps'}));
```

```
% Bar Graph
```

```
figure;
```

```
bar(1:Users, Rate);  
xlabel('Users');  
ylabel('Data Rate (kbps)');  
title('FDMA Data Rate for Each User');  
grid on;  
output:
```

User Data Rate (kbps)	
User	DataRate_kbps
1	200
2	200
3	200



Objective 3:

To evaluate the communication quality of the FDMA system by calculating the Signal-to-

Noise Ratio (SNR) for different users.

Output Parameter: Signal-to-Noise Ratio (SNR) (dB)

clc;

clear;

close all;

% Signal Power

Ps = 1; % Signal Power (W)

% Noise Power for Each User

Pn = [0.1 0.05 0.01]; % Noise Power (W)

% User Numbers

Users = 1:3;

% Calculate SNR in dB

SNR = 10 * log10(Ps ./ Pn);

% Display Results

```
disp('User  Noise Power (W)  SNR (dB)');  
disp([Users' Pn' SNR']);
```

```
% Plot SNR
```

```
figure;
```

```
bar(Users, SNR);
```

```
xlabel('Users');
```

```
ylabel('SNR (dB)');
```

```
title('Signal-to-Noise Ratio of FDMA Users');
```

```
grid on;
```

